

Quiz and Coding Exercise

Quiz:

1. What does the `break` statement do?

- a) Skips the current iteration of a loop
- b) Exits the loop immediately
- c) Continues to the next iteration of a loop
- d) Does nothing

2. When is the `continue` statement used?

- a) To exit a loop prematurely
- b) To skip the rest of the code for the current iteration and move to the next
- c) As a placeholder statement
- d) To iterate through a list

3. What is the purpose of the `pass` statement?

- a) Exits a loop
- b) Skips the current iteration of a loop
- c) Acts as a null operation, doing nothing
- d) Skips the rest of the code for the current iteration

Coding Exercise:

Problem 1: Using `break` in a While Loop

Write a Python program that takes a list of numbers as input `numbers = [25, 30, 20, 40, 15, 25]` and prints the sum of the numbers. However, if the sum exceeds 100, stop adding numbers and print "Sum exceeded 100".

Problem 2: Using `continue` in a For Loop

Write a Python script that uses a for loop to iterate through numbers from 1 to 600. Print only the odd numbers, skipping the even ones using the `continue` statement.

Problem 3: Using `pass` in Conditional Statements

Write a Python script that checks if a number is even or odd. If the number is even, print "Even"; if odd, do nothing (use the `pass` statement).

Problem 4: Combining Transfer Statements

Write a Python script that iterates through a list of words. If the word is "break," exit the loop using the `break` statement. If the word is "skip," skip the rest of the code for the current iteration using the `continue` statement. For any other word, print the word.